Priority 1 - Causal Data Generator Validation Summary

InGen Dynamics | Predictive Maintenance for Robot Fleets

Branch: fix/causal-data-generator | Commit: 95489cc | Status: Initial validation complete

1. Objective

Improve the causal consistency of the synthetic robot fleet dataset: completed mission usage drives component wear, wear affects fault risk, and maintenance recovery occurs only after repair completion.

Remove retrospective bearing-fault precursor injection and validate temporal consistency across missions, telemetry, faults, and maintenance.

2. Key Implementation Changes

Chronological simulation: completed missions contribute to operating hours and component degradation; the resulting health state influences subsequent fault probabilities.

Bearing signal: motor current increases with simulated bearing degradation. Bearing precursor signals are not inserted retrospectively based on known future faults.

Maintenance recovery: component wear is reduced only at the first telemetry step after the repair end time.

Mission consistency: mission_status distinguishes completed, cancelled_maintenance, and not_observed missions; cancelled missions do not contribute simulated usage.

3. Validation Results

Metric	Result
Robots	50
Missions	46,327
Telemetry rows	109,324
Fault events	273
Maintenance events	133
Completed missions	46,048
Cancelled due to maintenance	269
Not observed	10
Faults during maintenance	0
Completed missions overlapping maintenance	0
Missing mission statuses	0
Retrospective bearing precursors	0

4. Maintenance Recovery Check

Across 80 eligible bearing maintenance events, 79 (98.75%) showed lower mean motor current in the three days after maintenance than in the three days before.

Mean current before repair: 4.943 A; after repair: 4.224 A; reduction: 0.719 A (approximately 14.5%). These observations are consistent with the intended recovery mechanism.

5. Historical Signal and Future Fault Risk

Prediction-time features used mean motor current over [T-7 days, T); the outcome was a bearing fault during [T, T+7 days). The check covered 26,602 robot-days.

High historical current was associated with approximately 3.93 times the future seven-day bearing fault rate of low historical current. This is a predictive association, not causal identification or a full downstream model leakage audit.

Historical current	Mean current	Future 7-day bearing fault rate
Low	4.220 A	1.838%
Medium	4.559 A	3.643%
High	5.140 A	7.229%

6. Implementation and Version Control

New script: scripts/W01_generate_fleet_causal.py. Original generator preserved as the baseline.

Default preview output: data/causal_preview/raw/ and data/causal_preview/sealed/. The preview directory is excluded through .gitignore.

Local branch: fix/causal-data-generator; local commit: 95489cc. The commit includes the new generator and .gitignore, not generated CSVs.

7. Conclusion and Remaining Checks

The initial generator repair and focused QA checks support the intended chronological usage–wear–fault–repair relationships and the internal consistency of completed missions and maintenance events.

Before downstream modeling, complete a full feature/label leakage audit, verify temporal train/test separation and data contracts, and rerun validation after any generator changes.